

- 5** Source S_1 , consisting of parallel light with wavelength 700 nm, is incident on a rectangular slit of width b , as shown in Fig. 5.1.

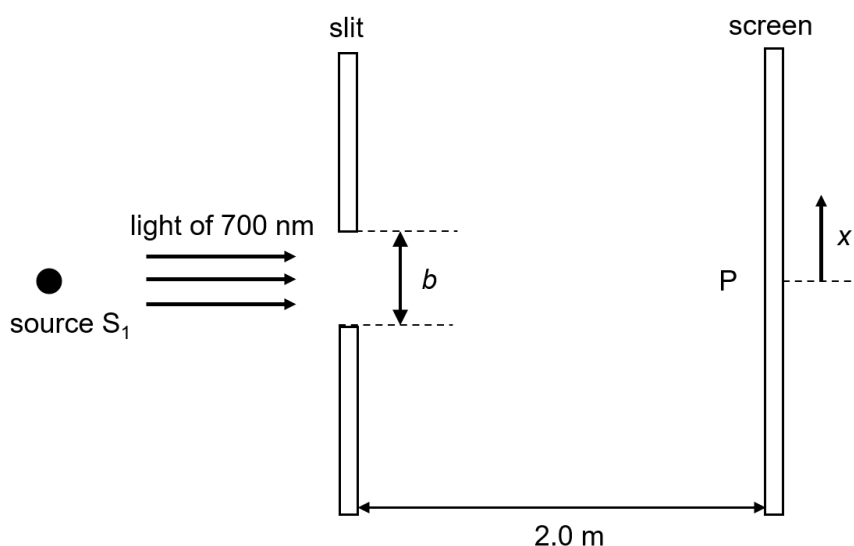


Fig. 5.1 (not to scale)

- (a) A central maxima is observed on the screen and its width is found to be 4.0 mm . Calculate the width b of the single slit.

Since θ is small, $\theta = \frac{\lambda}{b} = \tan \theta = \frac{x}{D}$

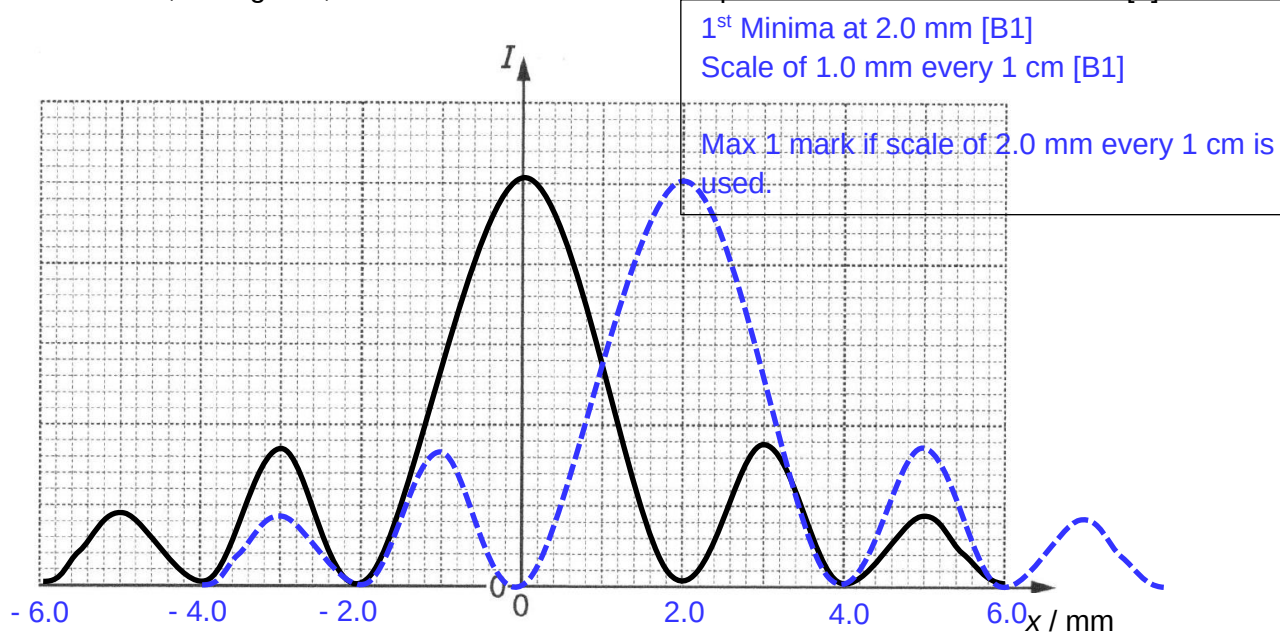
$$\frac{700 \times 10^{-9}}{b} = \frac{0.002}{2.0} \text{ [C1]}$$

$$b = 7.0 \times 10^{-4} \text{ m [A1]}$$

Comment

Poorly done. A number of candidates do not remember that x is half the width of the central maxima.

- (b) Fig. 5.2 shows the variation with distance x from P of the intensity I of the red light on the screen. Label, on Fig. 5.2, the values of the six x -intercepts. [2]



Comment

Poorly done. Candidates equated slit width with width of central maxima which is conceptually wrong. Uneven scaling of x axis is observed. Many missed this question.

- (c) A

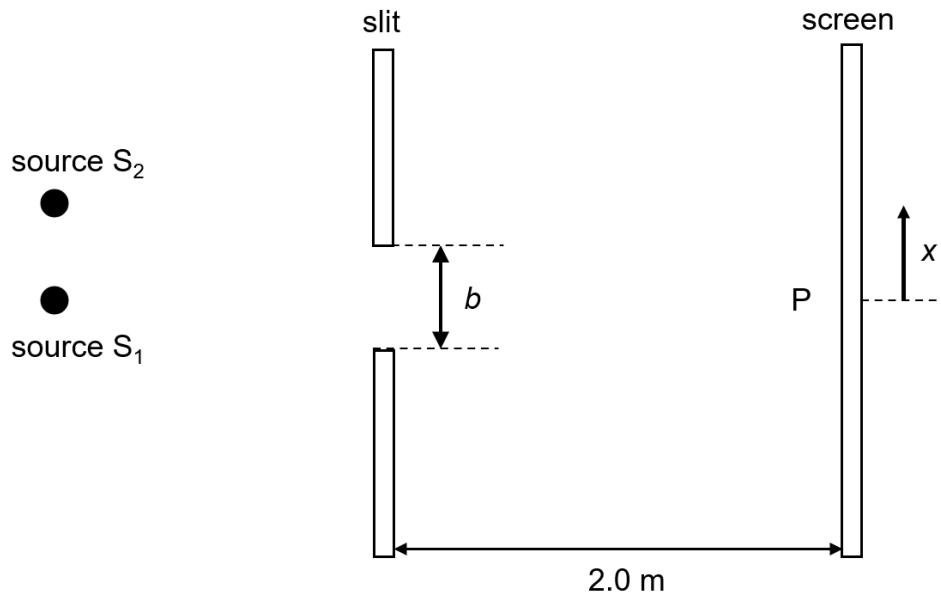


Fig. 5.3 (not to scale)

Sketch, on Fig. 5.2, the variation with distance x from P of the intensity I of the second source S_2 when Rayleigh criterion is satisfied. [2]

[Total: 9]

Central maxima of new diffraction envelop at first minima of original diffraction envelop [B1]

1st Minima of new diffraction envelop at central maxima AND 2nd minima of original diffraction envelop [B1] – rephrase using terms in Rayleigh's criterion.

Comment

Very poorly done. Candidates did not read question and drew on Fig. 5.3 instead. Curve was drawn without regards to Rayleigh Criterion. Awkwardly shaped graphs and unsymmetrical graphs are observed. Many missed this question.

Comment

Very poorly done. Confusion between magnetic flux density, magnetic flux and magnetic flux linkage is common. Magnetic field is insufficient as it describes a region and not quantify the strength.